Songhai "Frank" Fan

/ˈsɒŋˌhaɪ/ /fæn/

AI Product Researcher

songhai.fan2022@gmail.com · +61 434 780 632 · Melbourne, Australia
github.com/SonghaiFan · songhai.one · linkedin.com/in/songhaifan

AI product researcher and prototyper with a PhD in Human-Centred Computing from Monash University and a master's background in Data Science. My data science training gave me a strong foundation in databases, modelling, and analytical problem-solving, while my PhD focused on building interactive systems, AI-assisted workflows, and AI prototypes that help people understand and work with complex information. I have 7 years of programming experience and 5 years of research experience across information visualisation, NLP-based graph modelling, LLM interaction and user studies.

PROFESSIONAL EXPERIENCE

2021-2026

PhD Researcher & AI Prototyper

Monash University - PhD Research Program

- Led a five-year PhD research program on how people understand complex text, narratives and AI-generated structure.
- Designed and built AI-assisted tools that turn long-form news into editable visual narrative graphs people can inspect and adjust.
- Ran a 50-participant user study showing visual AI interfaces helped people complete analysis tasks 15-51% faster than text-only reading.

2022-2025

AI Visualisation Research Collaborator

Monash University / External research partners

- Built data visualisation and text-analysis prototypes for public-interest projects in policy, media and climate communication.
- Used AI-assisted extraction, summarisation and clustering to help domain experts inspect patterns across large text collections.

2021-PRESENT

Data Visualisation Educator & Project Mentor

Monash University

- Taught data visualisation, HCI, D3, data wrangling and data science to undergraduate and postgraduate students.
- Mentored 200+ students through visualisation, UX and data product projects, with a focus on clear problem framing and evidence-based design.

SELECTED PROJECTS

AI-ASSISTED TOOL / DATA VISUALISATION / USER STUDY

Human-AI Narrative Analysis Tool

End-to-end prototype for helping people inspect how news narratives change over time, with AI generation plus manual user control.

- Designed the Time-Track Narrative Graph model and a visual interface for analysing time-dependent news narratives.
- Evaluated the system with 50 participants; visual interfaces improved task speed by 15-51% compared with text-only reading.

VOICE AGENT / LOCAL AI / SERVICE PROTOTYPE

Fabula AI Storytelling Companion

Voice-driven AI prototype that helps older adults organise life stories through guided conversation and a visual storyboard.

- Designed a conversational flow that supports gentle, structured interviews rather than open-ended chatbot interaction.
- Built a local AI prototype with persona control, voice interaction and automated setup.

HUMAN-AI REASONING / GRAPH INTERFACE / BENCHMARK

Causal Reasoning Explorer

Tools for helping people inspect causal assumptions in policy models and sentence-level causal networks extracted from text.

- Designed interfaces for inspecting causal assumptions in policy models and causal networks extracted from text.
- Implemented topic clustering, centrality analysis, filtering and search for sentence-level causal graph exploration.

STAKEHOLDER PROTOTYPE / AI TEXT ANALYSIS / VISUAL ANALYTICS

Australia Discourse Explorer

Interactive visual analytics prototypes for helping researchers trace storylines across opinion articles and parliamentary speeches.

- Built visual analytics prototypes for a corpus of 180 million words from media and parliamentary text.
- Helped domain experts trace storylines across 23 disadvantaged groups and 33 social issues.

OPEN-SOURCE TOOL / D3 / REACT / COMMUNICATION DESIGN

ScrollyLite

Reusable scrollytelling library for turning complex data into guided, interactive explanations.

- Built public-facing data stories with domain experts in disadvantage policy and climate communication.
- Turned repeated design patterns into a reusable framework for step-based data storytelling.

EMBEDDING EXPLORATION / MULTIMODAL DATA / 3D PROTOTYPE

Music Topic Explorer

Interactive prototype for browsing musical theme-and-variation structure through annotated audio segments and embeddings.

- Curated a dataset of 692 annotated theme-and-variations recordings with 7,400+ verified audio segments.
- Combined handcrafted audio features and MERT embeddings for browsing similarity between musical variations.

CORE CAPABILITIES

Human-AI interaction, HCI, information visualisation and AI product research

7 years programming with React, D3, TypeScript, Python and SQL

AI-assisted development: prompt engineering, agentic workflows and evaluation harnesses

User-centred evaluation, controlled studies, interviews, mixed methods and usability testing

Clear communication with engineers, designers, researchers and domain experts

PRODUCT LENS

AI product discovery

AI prototyping

User-centred evaluation

Clear communication

RESEARCH EVIDENCE

IEEE VIS
2025 BEST SHORT PAPER
Out of the Loop

NAACL
2025
ACCESS: A Benchmark for Abstract Causal Event Discovery and Reasoning

TVCG
UNDER REVIEW
A Directed Graph Model and Experimental Framework for Time-Dependent Text Visualisation

EUROVIS
2026 UNDER REVIEW
Understanding Narrative in News: Visualisation Support for Explanatory Reporting

EDUCATION

PhD, Human-Centred Computing
Monash University, 2021-2026
Narrative visualisation / AI text visualisation / Human-centred data interfaces

Master of Data Science, Distinction
Monash University, 2019-2021
Business intelligence / Data exploration and visualisation / Statistical modelling / Big data processing / Data wrangling / Data warehousing

Bachelor of Engineering, Honours
Monash University / Central South University, 2014-2018

WORKING STYLE

Start with the human workflow, then decide where AI belongs

Build small working prototypes and test them with real tasks

Keep users in control instead of hiding everything behind automation

TOOLBOX

Prompt engineering, agentic coding, AI-assisted development

React, D3, TypeScript, Python, SQL

User studies, mixed methods, usability testing

FOCUS AREAS

Human-AI interaction

AI product prototyping

Data visualisation

Research-to-product translation